P1 Anisotropic harmonic oscillator

$$V = \frac{1}{2}\mu\left(\omega_1^2(x^2+y^2) + \omega_3^2 z^2\right)$$

$\omega_1 < \omega_3 < 2\omega_1$.

$$H = \underbrace{\frac{p_x^2}{2\mu} + \frac{1}{2}\mu\omega_1^2 x^2}_{H_x} + \underbrace{\frac{p_y^2}{2\mu} + \frac{1}{2}\mu\omega_1^2 y^2}_{H_y} + \underbrace{\frac{p_z^2}{2\mu} + \frac{1}{2}\mu\omega_3^2 z^2}_{H_z}$$

Since $[H_x, H_y] = [H_x, H_z] = [H_y, H_z] = 0$, the energy eigenvalues are given by the <u>sum</u> of those for one-dimensional oscillators H_x, H_y, and H_z.

$$\Rightarrow E = \hbar\omega_1\left(n_x + \tfrac{1}{2}\right) + \hbar\omega_1\left(n_y + \tfrac{1}{2}\right) + \hbar\omega_3\left(n_z + \tfrac{1}{2}\right)$$

$$= \hbar\omega_1\left(n_x + n_y + 1\right) + \hbar\omega_3\left(n_z + \tfrac{1}{2}\right)$$

where $n_x, n_y, n_z = 0, 1, 2 \ldots$

<u>Lowest</u> energy: $n_x = n_y = n_z = 0$

$$\Rightarrow E_{000} = \hbar\omega_1 + \frac{1}{2}\hbar\omega_3 = E_0$$

<u>First excited state</u>: $\begin{cases} n_x = 1, & n_y = 0, \quad n_z = 0 \\ n_x = 0, & n_y = 1, \quad n_z = 0 \end{cases}$

$$E_1 = E_{010} = E_{100} = \hbar\omega_1(1+1) + \hbar\omega_3\frac{1}{2} = \hbar\left(2\omega_1 + \frac{\omega_3}{2}\right)$$

is <u>two-fold</u> degenerate

<u>2nd excited state</u>: E_{001} for $n_x = n_y = 0$, $n_z = 1$

$$E_2 = E_{001} = \hbar\omega_1 + \hbar\omega_3\left(1 + \tfrac{1}{2}\right) = \hbar\left(\omega_1 + \tfrac{3}{2}\omega_3\right)$$

is not degenerate.

P2

$$Y_{2,1}(\theta,\phi) = -\sqrt{\frac{15}{8\pi}}\,\sin\theta\cos\theta\, e^{i\phi}$$

$$\hat{L}_+ = \hbar e^{i\phi}\left(\frac{\partial}{\partial\theta} + i\cot\theta\frac{\partial}{\partial\phi}\right)$$

$$\hat{L}_+ Y_{2,1}(\theta,\phi) = -\sqrt{\frac{15}{8\pi}}\,\hbar\, e^{i\phi}\left(\frac{\partial}{\partial\theta} + i\cot\theta\frac{\partial}{\partial\phi}\right) e^{i\phi}\sin\theta\cos\theta =$$

$$= -\sqrt{\frac{15}{8\pi}}\,\hbar\left(e^{i2\phi}\frac{\partial}{\partial\theta}\sin\theta\cos\theta + i e^{i\phi}\cot\theta\,\sin\theta\cos\theta\,\frac{\partial e^{i\phi}}{\partial\phi}\right) =$$

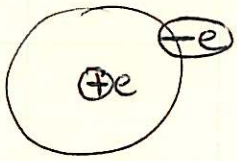
$$= -\sqrt{\frac{15}{8\pi}}\,\hbar\, e^{i2\phi}\left(\cos^2\theta - \sin^2\theta + i^2\cos^2\theta\right) = \sqrt{\frac{15}{8\pi}}\,\hbar\, e^{i2\phi}\sin^2\theta$$

On the other hand, $\hat{L}_+ Y_{2,1}(\theta,\phi) = \hbar\underbrace{\sqrt{l(l+1)-m(m+1)}}_{l=2,\ m=1} Y_{2,2}(\theta,\phi) =$

$$= \hbar\sqrt{2\cdot 3 - 1\cdot 2}\, Y_{2,2} = 2\hbar\, Y_{2,2}$$

$$\Rightarrow Y_{2,2}(\theta,\phi) = \frac{1}{2\hbar}\sqrt{\frac{15}{8\pi}}\,\hbar\, e^{i2\phi}\sin^2\theta = \sqrt{\frac{15}{32\pi}}\, e^{i2\phi}\sin^2\theta$$

P3



$$\psi_{100}(r,\theta,\phi) = \underbrace{\frac{2}{a_0^{3/2}}\, e^{-r/a_0}}_{R_{10}(r)} \cdot \underbrace{\frac{1}{\sqrt{4\pi}}}_{Y_{0,0}(\theta,\phi)}$$

(a)

$$P = \int d^3\vec{r}\,|\psi_{100}|^2 = \int_0^b dr\, r^2\, R_{10}^2(r)$$

$$\Rightarrow P = \frac{4}{a_0^3}\int_0^b dr\, r^2 e^{-2r/a_0} = \left(r = \frac{a_0}{2}x\right) =$$

$$= \frac{4}{a_0^3}\left(\frac{a_0}{2}\right)^3\int_0^{2b/a_0} dx\cdot x^2 e^{-x} = \frac{1}{2}\int_0^{\varepsilon} dx\, x^2 e^{-x}$$

$$\varepsilon = 2b/a_0$$

$$\int_0^{\varepsilon} dx\, x^2 e^{-x} = \int_0^{\varepsilon} d(-e^{-x})\, x^2 = -x^2 e^{-x}\Big|_0^{\varepsilon} - \int_0^{\varepsilon} d(x^2)\,(-e^{-x})$$

$$= -\varepsilon^2 e^{-\varepsilon} + 2\int_0^{\varepsilon} dx\cdot x e^{-x} = -\varepsilon^2 e^{-\varepsilon} + 2\int_0^{\varepsilon} d(-e^{-x})\, x =$$

$$= -\varepsilon^2 e^{-\varepsilon} + 2\left[-x e^{-x}\Big|_0^{\varepsilon} + \int_0^{\varepsilon} dx\, e^{-x}\right] =$$

$$= -\varepsilon^2 e^{-\varepsilon} - 2\varepsilon e^{-\varepsilon} + 2\left(1 - e^{-\varepsilon}\right)$$

$$\Rightarrow P = \frac{1}{2}\, 2\left(1 - e^{-\varepsilon} - \varepsilon e^{-\varepsilon} - \frac{\varepsilon^2}{2} e^{-\varepsilon}\right) =$$

$$= 1 - e^{-\varepsilon}\left(1 + \varepsilon + \frac{\varepsilon^2}{2}\right) = \left(\varepsilon = 2b/a_0\right) =$$

$$= 1 - e^{-2b/a_0}\left(1 + \frac{2b}{a_0} + \frac{2b^2}{a_0^2}\right)$$

(b) Expand P in powers of $\varepsilon = 2b/a_0$:

$$P \cong 1 - \left(1 - \varepsilon + \frac{1}{2}\varepsilon^2 - \frac{1}{6}\varepsilon^3\right)\left(1 + \varepsilon + \frac{1}{2}\varepsilon^2\right) =$$

$$= 1 - \left(1 + \underline{\varepsilon} + \underline{\underline{\frac{1}{2}\varepsilon^2}} - \underline{\varepsilon} - \underline{\underline{\varepsilon^2}} - \frac{1}{2}\varepsilon^3 + \underline{\underline{\frac{1}{2}\varepsilon^2}} + \frac{1}{2}\varepsilon^3 - \frac{1}{6}\varepsilon^3\right)$$

$$= 1 - \left(1 - \frac{1}{6}\varepsilon^3\right) = \frac{1}{6}\varepsilon^3$$

$$\Rightarrow P \approx \frac{1}{6}\left(\frac{2b}{a_0}\right)^3 = \frac{8b^3}{6a_0^3} = \frac{4}{3}\left(\frac{b}{a_0}\right)^3$$

(c) Assume that $\psi(r \sim b) = \psi(0)$, then

$$P = \left(\frac{4\pi b^3}{3}\right) \cdot |\psi(0)|^2 = \frac{4\pi}{3} b^3 \cdot \left(\frac{2}{a_0^{3/2}} \cdot \frac{1}{\sqrt{4\pi}}\right)^2 = \frac{4\pi b^3}{3} \cdot \frac{1}{\pi a_0^3}$$

↳ volume of the sphere

$$\Rightarrow P = \frac{4b^3}{3a_0^2} \quad \text{as in (b).}$$

(d) $b = 10^{-13}$ cm, $a_0 = 0.5 \cdot 10^{-8}$ cm $\Rightarrow P = \frac{4}{3}\left(\frac{10^{-13}}{0.5 \cdot 10^{-8}}\right)^3 =$

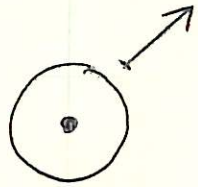
$$= \frac{4}{3}\, 2^3 \left(10^{-5}\right)^3 \simeq 10^{-14} \ll 1$$

$\Rightarrow$ expansion in $\varepsilon = \frac{2b}{a_0} \cong 4 \cdot 10^{-5}$ is well justified

(e) Electric field (from Gauss's law) of $(+e)$ sphere

$$E = \begin{cases} \frac{er}{b^3} & r < b \\ \frac{e}{r^2} & r > b \end{cases}, \quad V(r) = -\int E\,dr - (-e) = \begin{cases} \frac{e^2}{2b}\left(\frac{r^2}{b^2} - 3\right), & r < b \\ -\frac{e^2}{r}, & r > b \end{cases}$$

Electric field



$$E \cdot 4\pi r^2 = \begin{cases} 4\pi e, & r > b \\ 4\pi \left(\frac{r}{b}\right)^3 e, & r < b \end{cases}$$

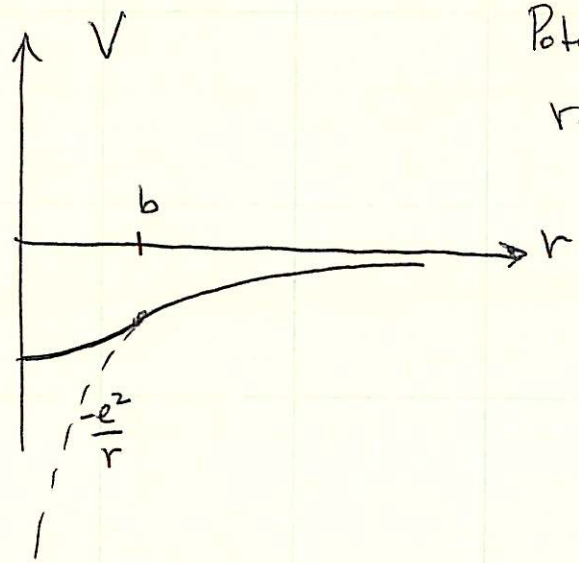
$$\Rightarrow E = \begin{cases} \frac{e}{r^2}, & r > b \\ \frac{er}{b^3}, & r < b \end{cases}$$

But electric potential $\varphi = -\int_{\infty}^{r} E\,dr =$

$$= -\int_{\infty}^{b} \frac{e}{r^2}\,dr - \int_{b}^{r} \frac{er}{b^3}\,dr = \frac{e}{b} - \frac{e}{2b^3}(r^2 - b^2)$$

Potential energy of an electron $V = -e\varphi = -\frac{e^2}{b} + \frac{e^2}{2b^3}(r^2 - b^2)$

$$= \frac{e^2}{2b^3}\left(r^2 - 3b^2\right) = \frac{e^2}{2b}\left(\frac{r^2}{b^2} - 3\right) \text{ for } r < b$$



Potential energy for $r > b$ remains the same:

$$V = -\frac{e^2}{r}, \quad r > b$$

P4 (a) $[\hat{q}_1, \hat{q}_2] = \frac{1}{2}[\hat{x}, \hat{x}] - \frac{a^2}{2\hbar}[\hat{x}, \hat{p}_y] + \frac{a^2}{2\hbar}[\hat{p}_y, \hat{x}] - \frac{a^4}{2\hbar^2}[\hat{p}_y, \hat{p}_y]$

$= 0$ (because every term is 0.)

$[\hat{p}_1, \hat{p}_2] = 0$ as well.

$$[\hat{q}_1, \hat{p}_1] = \frac{1}{2}\left[\hat{x} + \frac{a^2}{\hbar}\hat{p}_y, \hat{p}_x - \frac{\hbar}{a^2}\hat{y}\right] =$$

$$= \frac{1}{2}\underbrace{[\hat{x}, \hat{p}_x]}_{= i\hbar} - \frac{1}{2}\frac{a^2}{\hbar}\frac{\hbar}{a^2}\underbrace{[\hat{p}_y, \hat{y}]}_{= -i\hbar} = \frac{1}{2}(\hbar - (-i\hbar)) = i\hbar.$$

$$[\hat{q}_2, \hat{p}_2] = \frac{1}{2}\left[\hat{x} - \frac{a^2}{\hbar}\hat{p}_y, \hat{p}_x + \frac{\hbar}{a^2}\hat{y}\right] = i\hbar$$

(b) $\hat{x} = \frac{1}{\sqrt{2}}(\hat{q}_1 + \hat{q}_2) \qquad \hat{y} = \frac{a^2}{\sqrt{2}\hbar}(\hat{p}_2 - \hat{p}_1)$

$\hat{p}_x = \frac{1}{\sqrt{2}}(\hat{p}_1 + \hat{p}_2) \qquad \hat{p}_y = \frac{\hbar}{\sqrt{2}a^2}(\hat{q}_1 - \hat{q}_2)$

$$\hat{L}_z = \hat{x}\hat{p}_y - \hat{y}\hat{p}_x = \frac{\hbar}{2a^2}(\hat{q}_1 + \hat{q}_2)(\hat{q}_1 - \hat{q}_2) -$$

$$- \frac{a^2}{\sqrt{2}\hbar}\frac{1}{\sqrt{2}}(\hat{p}_2 - \hat{p}_1)(\hat{p}_1 + \hat{p}_2) = \frac{\hbar}{2a^2}(\hat{q}_1^2 \underbrace{- \hat{q}_1\hat{q}_2 + \hat{q}_2\hat{q}_1}_{=0} - \hat{q}_2^2)$$

$$+ \frac{a^2}{2\hbar}(\hat{p}_1^2 \underbrace{+ \hat{p}_1\hat{p}_2 - \hat{p}_2\hat{p}_1}_{=0} - \hat{p}_2^2)$$

$$\Rightarrow \hat{L}_z = \frac{\hbar}{2a^2}(\hat{q}_1^2 - \hat{q}_2^2) + \frac{a^2}{2\hbar}(\hat{p}_1^2 - \hat{p}_2^2)$$

(c) Let $\hat{H}_1 = \frac{\hbar}{2a^2}\hat{q}_1^2 + \frac{a^2}{2\hbar}\hat{p}_1^2$

$\hat{H}_2 = \frac{\hbar}{2a^2}\hat{q}_2^2 + \frac{a^2}{2\hbar}\hat{p}_2^2$

Denote $m = \frac{\hbar}{a^2} \Rightarrow \begin{cases} \hat{H}_1 = \frac{1}{2m}\hat{q}_1^2 + \frac{1}{2}m\hat{p}_1^2 \\ \hat{H}_2 = \frac{1}{2m}\hat{q}_2^2 + \frac{1}{2}m\hat{p}_2^2 \end{cases}$

$\Rightarrow$ $\hat{H}_{1,2}$ describe harmonic oscillators of frequency $\omega = 1$

Note that because of (a) $[\hat{H}_1, \hat{H}_2] = 0$.

(d) Eigenvalues of $\hat{H}_1$ are E_1 such that

$$\hat{H}_1 |\psi\rangle = E_1 |\psi\rangle = \hbar\omega \left(n_1 + \tfrac{1}{2}\right) |\psi\rangle = (\omega = 1) =$$
$$= \hbar \left(n_1 + \tfrac{1}{2}\right) |\psi\rangle$$

and $\hat{H}_2 |\psi\rangle = E_2 |\psi\rangle = \hbar\omega \left(n_2 + \tfrac{1}{2}\right) |\psi\rangle = \hbar \left(n_2 + \tfrac{1}{2}\right) |\psi\rangle$

But the eigenstate $|\psi\rangle$ is the same for $\hat{H}_1$ and $\hat{H}_2$ because they commute.

Thus $\hat{L}_z |\psi\rangle = (\hat{H}_1 - \hat{H}_2) |\psi\rangle = \hbar \left(n_1 + \tfrac{1}{2} - n_2 - \tfrac{1}{2}\right) |\psi\rangle$

$$= \hbar (n_1 - n_2) |\psi\rangle = \hbar m |\psi\rangle$$

where $m = n_1 - n_2 = \underline{\underline{\text{Integer}}}$ because n_1 and n_2 are integers